



Personal Info



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27/11/1996



Italian



Languages

- English – B2
- Spanish – A2



Sports

- Inline / ice Speed Skating
- Gym
- Cycling
- Running



Skills

Programming languages:

- C / C++
- Python
- HTML / CSS / JavaScript
- Matlab / Simulink / Stateflow

Version Control System:

- Git
- TortoiseHg

AI:

- Claude / Claude Code
- Perplexity
- ChatGPT
- Gemini

IDE:

- Visual Studio Code
- IAR
- Eclipse
- MCUXpresso
- ESP-IDF

RTOS:

- FreeRTOS
- embOS
- ThreadX

Samuel Lazzaro

Software / Firmware Developer

I'm a Software / Firmware Developer with a passion for developing cutting-edge technologies. Driven by curiosity and a relentless desire to learn, I thrive on turning challenges into opportunities to create value and support those around me.

Work Experience

April 2024 – up to present **Firmware Engineer**

Carlo Gavazzi Controls – Lainate (MI) – Italy

I am responsible for the Web Server of Energy Meters on an esp32-mini-1, covering both frontend and backend to specifications from sales managers.

For the frontend, I developed using HTML, CSS, and JavaScript without any framework to ensure minimal resource usage, as the available memory on the esp32 was highly constrained. The primary goal was to keep the total size of the web files below 500 kB, which I successfully achieved by optimizing the files to approximately 100 kB. I also implemented a responsive design for mobile phones, tablets, and laptops.

For the backend, I developed the server logic in C using the ESP-IDF framework and Espressif libraries. A key challenge was managing heap memory consumption due to the overhead introduced by the HTTPS server for each active connection. To address this, I implemented a mechanism to limit the number of concurrent users, ensuring memory constraints were respected while maintaining system stability and performance.

I also set up the device network connectivity (Ethernet, Wi-Fi STA and SoftAP) using the ESP-IDF framework.

To enhance reliability and maintainability, I integrated automated tests using Python Behave with Selenium or Playwright scripts to simulate user interactions.

Additionally, I developed other project modules (alarms, datalogger, DMD calculation, network interface management) and their unit tests, verifying each module under different conditions.

IDEs used: IAR Embedded Workbench, Visual Studio Code with ESP-IDF extensions. MCUs used: ESP32, STM32.

Protocols used: Modbus TCP/RTU, LoRaWAN, HTTPS/TLS, REST API, UART, SPI, I2C.

November 2021 – March 2024

Firmware Development Junior Specialist

Custom SpA – Gallarate (VA) – Italy

As a firmware developer, I worked on electronic cash registers and fiscal printers marketed in both Italy and Germany. Each project required strict adherence to tax regulations while addressing customer demands for reliable, fast, and versatile devices.

The development was carried out in C/C++, using IDEs such as HEW and MCUXpresso. I used both embOS and Azure ThreadX as RTOS (Real-Time Operating Systems) and I integrated SQLite to handle relational databases efficiently.

Real-time debugging was conducted using a Renesas emulator or an NXP demo-board, enabling the accurate validation and fine-tuning of the application code in various test environments.

January 2021 – November 2021

Junior Software Developer

ALTEN Italia – Leonardo Helicopters – Cascina Costa (VA) – Italy

As a Junior Software Developer, I contributed to the development of the Large Area Display (LAD) graphical interface and operational logic for the AW249 military helicopter, designed by Leonardo Helicopters for the Italian Army. This position was carried out as a consultant for ALTEN Italia.

The graphical interface was developed using VAPS XT (Presagis), ensuring compliance with aviation standards and requirements, while the system's working logic was modeled using MATLAB/Simulink.

MCUs:

- ESP32
- STM32
- NXP
- Renesas

Communication protocols:

- Modbus TCP/RTU, LoRaWAN
- UART, SPI, I2C
- HTTPS/TLS, REST API

Other:

- GitHub
- Sourcetree (Atlassian)
- Excel
- VAPS XT



Personal projects

- www.centroesteticomaristella.it

This is the first website I developed for a beauty salon.

- www.euroskatingcardano2026.it

Website that I developed for the European Speed Skating Championships 2026.

- www.cardanoskating.it

Website that I'm developing for my speed skating team.



Certificates

(MCU1) Learn bare metal driver development using Embedded C: Writing drivers for STM32 GPIO,I2C,SPI,USART from scratch

- Date of issue: in progress
- Credential link: <https://www.udemy.com/course/mastering-microcontroller-with-peripheral-driver-development/?couponCode=KEEPEARNING>

Succeed in the Age of AI, Udemy

- Date of issue: January 2026
- Credential link: <https://www.udemy.com/certificate/UC-954008e6-921f-40b2-b1af-a4493a38dff2/>

The Complete 2024 Web Development Bootcamp, Udemy

- Date of issue: September 2024
- Credential link: <https://www.udemy.com/certificate/UC-858c013a-dc13-42d2-9add-5571df2d3e2d/>

Python 3.11 Complete Guide: from beginner to advance

- Date of issue: January 2022
- Credential link: <https://www.udemy.com/certificate/UC-b38135e4-5ff8-4b9f-87c1-90da7794b21a/>

Learn C++ Programming – Beginner to Advance – Deep Dive in C++

- Date of issue: July 2021
- Credential link: <https://www.udemy.com/certificate/UC-aac84e12-e43f-4c85-8757-fda7579d0715/>

For some software components, I implemented features using C/C++, focusing on efficiency and maintainability.

November 2020 – January 2021

Internship

ALTEN Italia – Milan (MI) – Italy

Software Design with Model Based approach, Software Development in Aerospace Field using C/C++ as programming languages.



Education

September 2018 – December 2020

Master's Degree in Aerospace Engineering – Aeromechanics and Systems

Politecnico di Torino, Turin

- **Thesis title:** Device for non-invasive Flight Test Instrumentation (FTI): hardware & software
- **Final grade:** 110/110 cum laude
- **Thesis description:** The thesis focused on developing a Flight Test Instrumentation (FTI) system. I began by conducting a market analysis to select suitable equipment, including oscilloscopes and sensors. Subsequently, I implemented C++ code for sensor fusion, enabling synchronized use of a camera and various sensors.

Two separate applications were developed for distinct purposes:

- Application 1: A GUI was created to display camera images alongside time-domain plots of sensor data (e.g., accelerations). The interface featured buttons to record camera feeds, save sensor data in defined time windows, and terminate the program. Additionally, a Fast Fourier Transform (FFT) feature allowed the analysis of accelerations in the frequency domain from the last time window.
- Application 2: A GUI was designed to plot time histories of angles, speeds, and accelerations detected by sensors. This application followed a Parameter Identification methodology to perform dynamic analysis of helicopters, estimating the natural frequency and damping of dynamic modes.

Both applications utilized the OpenCV library for computer vision tasks and the sensor's SDK for data acquisition and processing, demonstrating an advanced integration of software tools within C++.

September 2015 – July 2018

Bachelor's Degree in Aerospace Engineering Percorso "Giovani Talenti"

Politecnico di Torino, Turin

- **Thesis title:** Use of the superelements technique in the FEM analysis of the S55 Replica aircraft
- **Final grade:** 110/110 cum laude

September 2010 – June 2015

High school diploma Maturità Scientifica

Liceo Scientifico Giuseppe Moscati, Grottaglie (TA)

- **Final grade:** 100/100